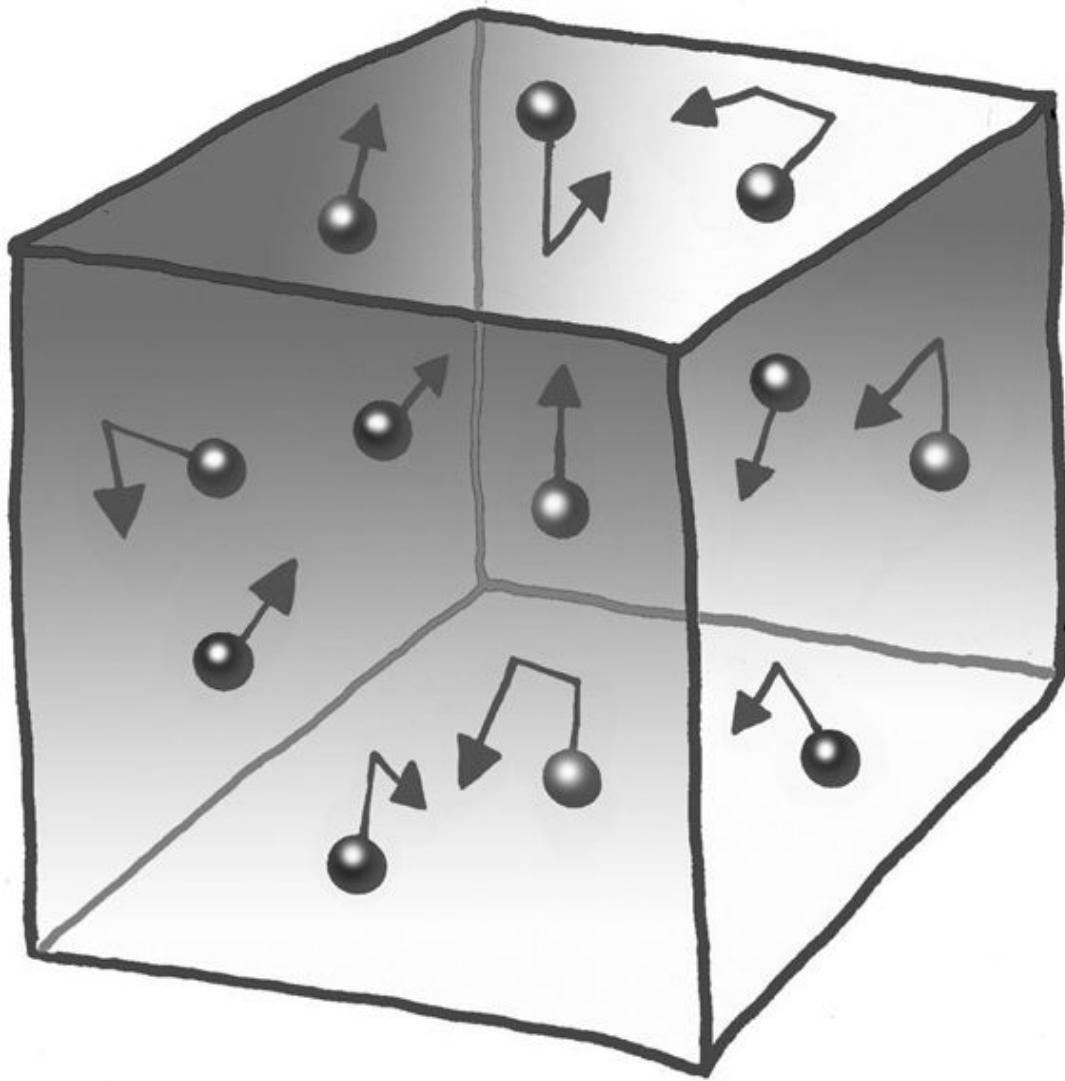


Democritus and Leucippus: atomism



A theory of matter proposed by Leucippus and his pupil Democritus was less influential at the time than the ‘four elements’ proposed by their contemporary Empedocles (see [The four elements](#)), but in retrospect seems closer to modern scientific understanding. They suggested that everything in the universe is composed of minute, unalterable and indivisible particles, which they called atoms (from the Greek *atomos*, uncuttable). These, they argued, are free to move through empty space, combining in constantly changing configurations.

The assertion there is such a thing as a void, an empty space, may be one reason these ideas were originally considered unacceptable. According to their theory, the number of atoms is infinite, and different kinds of atoms with different characteristics determine the properties of the substance they form together. Because the atoms are indestructible, when a substance, or even a human body, decays, its atoms are dispersed and reconstituted in another form.